

**Assignment 3**

TU Delft, ME, DCSC, Spring 2025

- The assignments are individual. You may consult and discuss with your colleagues, but the answers you provide, the code you use, and the report you hand in must be independent.
- Provide detailed answers, describing the steps you followed. Only end results of calculations are not sufficient.
- Provide a clear report as a PDF file, typed, preferably using L<sup>A</sup>T<sub>E</sub>X.
- Submit the report digitally via Brightspace before the deadline: 9:00, June 20, 2025.

3p

**Problem 1**

Consider the convex quadratic optimization problem

$$\begin{aligned} & \underset{x}{\text{minimize}} && f(x) \\ & \text{subject to} && x \in \mathcal{X} \end{aligned}$$

where  $f(x) = \frac{1}{2}x^\top Hx + c^\top x$ ,  $H \geq 0$ , is a convex quadratic function and  $\mathcal{X}$  is a convex set.

Show that  $x^*$  is an optimal solution if and only if

$$(z - x^*)^\top (-\nabla f(x^*)) \leq 0, \quad \forall z \in \mathcal{X}$$

Figure 1a depicts this condition for  $x \in \mathbb{R}^2$ . This condition motivates defining the so-called normal cone to  $\mathcal{X}$  at  $x^*$  as follows

$$\mathcal{N}(\mathcal{X}, x^*) = \{y \mid (z - x^*)^\top (y - x^*) \leq 0, \quad \forall z \in \mathcal{X}\}$$

The optimality condition then can be stated equivalently as:  $x^*$  is an optimal point if and only if the negative gradient is in the normal cone to  $\mathcal{X}$  at  $x^*$ , i.e.,

$$-\nabla f(x^*) \in \mathcal{N}(\mathcal{X}, x^*)$$

This condition and the normal cone are depicted in Figure 1b.

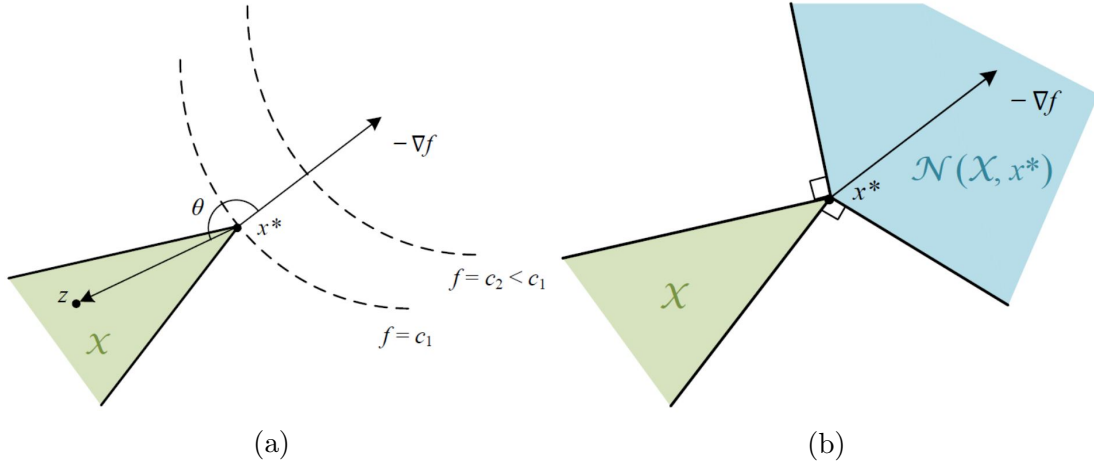


Figure 1: (a) Optimality of  $x^*$  means the angle  $\theta$  between  $-\nabla f$  and any point  $z$  in the feasible region must be greater than  $90^\circ$  and less than  $270^\circ$ . (b) The same result restated:  $x^*$  is optimal if and only if the negative gradient is in the normal cone to the feasible region at  $x^*$ , i.e.,  $-\nabla f(x^*) \in \mathcal{N}(\mathcal{X}, x^*)$

3p

## Problem 2

Consider the positive definite quadratic function partitioned into two sets of variables

$$V(u) = \frac{1}{2} u^\top H u + c^\top u + d$$

$$V(u_1, u_2) = \frac{1}{2} \begin{pmatrix} u_1^\top & u_2^\top \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} c_1^\top & c_2^\top \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + d$$

in which  $H > 0$ . Imagine we wish to optimize this function by the following procedure: First optimize over the  $u_1$  variables holding  $u_2$  fixed, and in parallel optimize over the  $u_2$  variables holding  $u_1$  fixed. Then create a convex combination of the two results as shown in Figure 2 for a scalar case.

Let us see if this procedure, while not necessarily efficient, is guaranteed to converge to the optimum.

1p

(a) Given an initial point  $(u_1^p, u_2^p)$ , show that the next iteration is

$$\begin{pmatrix} u_1^{p+1} \\ u_2^{p+1} \end{pmatrix} = w_1 \begin{pmatrix} u_1^{p+}(u_2^p) \\ u_2^p \end{pmatrix} + w_2 \begin{pmatrix} u_1^p \\ u_2^{p+}(u_1^p) \end{pmatrix}, \quad 0 \leq w_1, w_2, \quad w_1 + w_2 = 1,$$

where

$$u_1^{p+}(u_2^p) = -H_{11}^{-1} (H_{12}u_2^p + c_1)$$

$$u_2^{p+}(u_1^p) = -H_{22}^{-1} (H_{21}u_1^p + c_2)$$

Show that the procedure can be summarized as

$$u^{p+1} = A u^p + b$$

in which the iteration matrix  $A$  and constant  $b$  are given by

$$A = \begin{pmatrix} w_2 I & -w_1 H_{11}^{-1} H_{12} \\ -w_2 H_{22}^{-1} H_{21} & w_1 I \end{pmatrix}, \quad b = \begin{pmatrix} -w_1 H_{11}^{-1} c_1 \\ -w_2 H_{22}^{-1} c_2 \end{pmatrix}$$

- 1p (b) Establish that the optimization procedure converges by showing the iteration matrix  $A$  is stable

$$|\text{eig}(A)| < 1$$

- 1p (c) Given that the iteration converges, show that it produces the same solution as

$$u^* = -H^{-1}c$$

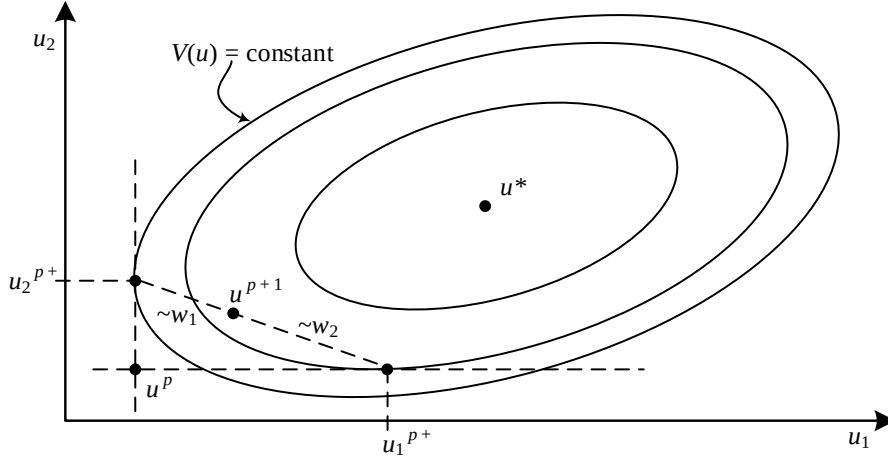


Figure 2: Optimizing a quadratic function in one set of variables at a time and then creating a convex combination for the next iterate.

### 2p Problem 3

Consider again the iteration defined in the previous problem. Prove that the cost function is monotonically decreasing when optimizing one variable at a time with the convex combination step

$$V(u^{p+1}) < V(u^p) \quad \forall u^p \neq -H^{-1}c$$

and show that the following expression gives the size of the decrease

$$V(u^{p+1}) - V(u^p) = -\frac{1}{2}(u^p - u^*)^\top P(u^p - u^*)$$

in which

$$P = HD^{-1}\tilde{H}D^{-1}H, \quad \tilde{H} = D - N, \quad H = D + N$$

$$D = \begin{pmatrix} w_1^{-1}H_{11} & 0 \\ 0 & w_2^{-1}H_{22} \end{pmatrix}, \quad N = \begin{pmatrix} -w_1^{-1}w_2H_{11} & H_{12} \\ H_{21} & -w_1w_2^{-1}H_{22} \end{pmatrix}$$

and  $u^* = -H^{-1}c$  is the optimum.

Hint: to simplify the algebra, first change coordinates and move the origin of the coordinate system to  $u^*$ .

Useful facts:

- If  $H$  is a positive definite, symmetric matrix partitioned in the following way, then

$$H = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} > 0 \quad \Rightarrow \quad \bar{H} = \begin{pmatrix} H_{11} & -H_{12} \\ -H_{21} & H_{22} \end{pmatrix} > 0.$$

- For any  $Q$  real symmetric and  $R$  real matrices:

$$Q > 0 \text{ and } R \text{ nonsingular} \quad \Rightarrow \quad R^\top Q R > 0.$$

1p

#### Problem 4

Consider the generalization of Problem 2 (without the convex combination step), in which we repeatedly decompose a problem into one-variable-at-a-time optimizations. For a three-variable problem we have the three optimizations

$$\begin{aligned} u_1^{p+1} &= \arg \min_{u_1} V(u_1, u_2^p, u_3^p) \\ u_2^{p+1} &= \arg \min_{u_2} V(u_1^p, u_2, u_3^p) \\ u_3^{p+1} &= \arg \min_{u_3} V(u_1^p, u_2^p, u_3) \end{aligned}$$

Is it true that

$$V(u_1^{p+1}, u_2^{p+1}, u_3^{p+1}) \leq V(u_1^p, u_2^p, u_3^p) \quad ?$$

Hint: you may wish to consider the example  $V(u) = \frac{1}{2}u^\top H u + c^\top u$ , where

$$H = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad u^p = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$